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ABSTRACT

FACULTY OF ENGINEERING AND PHYSICAL SCIENCES
ELECTRONICS AND COMPUTER SCIENCE

Bachelor of Science

Flood Inundation Maps In Zimbabwe: Intelligent Modelling Using Python

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Flooding is a frequent natural disaster that causes thousands of losses of lives every year. Flood Mapping is a key tool for mitigating the flood risk. The Flood Mapping topic is slowly diminishing in countries where the areal representation data of rivers is limited. This paper proposed a framework for filling the gap of the limited data availability by using a hydrologic model called PIHM to calculate the change in water depth and then feeding this to machine Learning Models; Random Forest and Multilayer Perceptron to forecast the change in water depth, taking Zimbabwe as a study area. Finally, flood maps were constructed using QGIS software. Random forest demonstrated good results and outperformed MLP. These two models were assessed using mean absolute error and mean square error. Finally, QGIS come to be a convenient tool for generating high-resolution flood maps, nevertheless, the accuracy of the maps is doubtful.